
An IoT-based smart parking management system

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Abstract: In this paper, the author implemented smart parking information management system based on IoT using ultrasonic parking sensor and Bluetooth beacon. Ultrasonic IoT sensors used for parking sensing are controlled by the Arduino board to collect parking sensing data, and the collected parking information is transmitted to the server in real time using the MQTT protocol. The server stores the parking information in the database and provides the vehicle driver with real-time parking status using the MQTT protocol. The driver starts the smartphone app in the parking lot and enters the parking lot. The smartphone automatically recognises which parking space using the Bluetooth beacon signal at the entrance of the parking lot and can confirm the parking information of the parking space to be parked at a glance by communicating with the server. The parking management system implemented in this study not only helps the driver to park the car but also uses the real-time parking information stored in the database to obtain marketing information such as hourly, daily or monthly number of visiting customers and average shopping time.

Keywords: IoT; smart parking; Bluetooth beacon; parking app; message queue telemetry transport; MQTT.

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Biographical notes: Inhwan Jung received his PhD in Information and Communication Engineering from KAIST in 2000. He is currently a Professor in Department of Computer Engineering at Hansung University, Seoul, South Korea. His major research interest is focused on topics related to IoT and mobile software.

1 Introduction

In modern society, finding vacant parking space when using a car in a large parking lot such as a department store is always a concern. In areas where large parking lots are already available, such as department stores and large discount stores, the number of available spaces is displayed at the entrance of the floor to facilitate parking. However, this method has no way for people to find out exactly where they are empty. To remedy

this problem LEDs are used to indicate empty parking spaces recently in the parking spaces. However, in this system, it is very uncomfortable to check the parking lot LED while driving. Recently, there has been a smart parking management research using IoT devices and message queue telemetry transport (MQTT) protocol (Khanna and Anand, 2016; Dhar and Gupta, 2016).

In this paper, an IoT-based smart parking guidance system is designed and implemented to help the driver to find the parking space more easily in a large parking lot such as a multiple story parking building. The author designed and implemented a smart parking system in which IoT sensors using an Arduino (<https://en.wikipedia.org/wiki/Arduino>) board and an ultrasonic sensor are installed on the ceiling of each parking space to check whether the vehicle is parked and to transmit parking information to the server. The Arduino board delivers the parking status collected from each parking sensor to the server using the MQTT protocol (<https://en.wikipedia.org/wiki/MQTT>). The server stores the collected parking space status information and provides the driver with a real-time parking status. A Bluetooth beacon (https://en.wikipedia.org/wiki/Bluetooth_low_energy_beacon) is installed at the entrance of the parking area to detect which parking space the vehicle enters.

In the implemented smart parking guidance system, the driver of the vehicle executes the parking guide application in advance and enters the parking space. The smartphone app automatically detects the position of the vehicle by sensing the signal of the Bluetooth beacon and receives the parking space information in real time by communicating with the server using the MQTT protocol and displays the parking space information. Therefore, the driver can conveniently go to an empty parking space quickly.

The parking information collected from the parking sensor can be used as various marketing information. In this paper, the author implemented web pages to display hourly, daily and monthly number of visiting customers and average shopping time.

The composition of this paper is as follows. Section 2 explains the research related to this paper. Section 3 explains the structure and function of smart parking guidance system. Section 4 describes the prototype of the parking information management system implemented and experiment contents. Section 5 shows server application that displays parking statistics. Finally, Section 6 presents conclusions and future research.

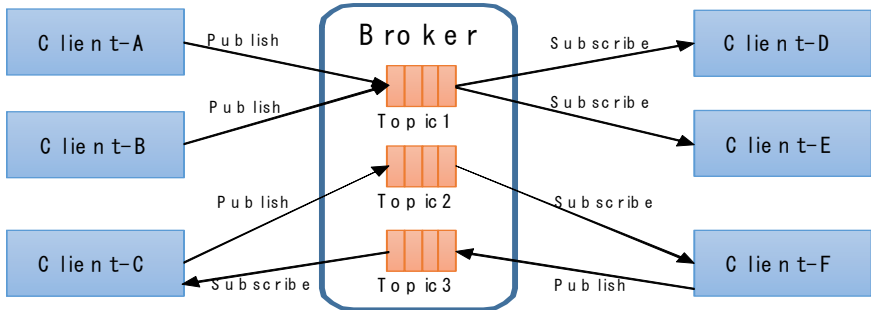
2 Related works

2.1 IoT data processing and MQTT

Various communication protocols such as constrained application protocol (CoAP) (<https://en.wikipedia.org/wiki/CoAP>), extensible messaging and presence protocol (XMPP) (<https://en.wikipedia.org/wiki/XMPP>) and MQTT have emerged to efficiently process IoT data. The MQTT protocol is a communication protocol for sending and receiving messages in a specific topic by using the *Subscribe* and *Publish* method. It provides an efficient communication environment for collecting data of a large number of IoT devices due to a short length of data and simple communication procedure (Lampkin et al., 2012; Aslava et al., 2015; Hwang et al., 2018).

The data flow using topic of MQTT is shown in Figure 1. The advantage of the MQTT structure is that multicasting or broadcasting can be performed using topic. As shown in Figure 1, client-A and client-B can transmit data to both client-D and client-E by publishing only once to topic 1.

Figure 1 MQTT topic and publish/subscribe data flow (see online version for colours)



2.2 *Arduino board*

Arduino board is a widely used device for IoT data acquisition. In this paper, the author use the Arduino MEGA board and add the Ethernet Shield for TCP communication with the server. The Arduino MEGA board has a total of 54 digital input/output pins and 16 analogue input pins. It connects the ultrasonic sensor using two digital pins for output and input. The output pins are used to emit ultrasonic pulse and input pins are used to receive pulses reflected from the object. Dividing the output and input time differences by the speed of the sound waves produces a distance. Ethernet Shield is used for MQTT communication using TCP connection.

Figure 2 Arduino MEGA board, Ethernet shield and ultrasonic sensor (see online version for colours)

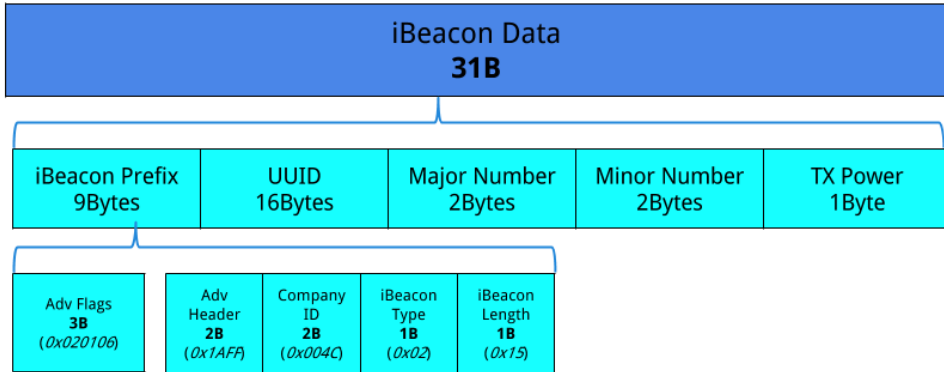


2.3 *Bluetooth beacon*

A beacon is a wireless communication device that can automatically recognise nearby smart devices and transmit necessary data. It is also called a Bluetooth beacon. While NFC (https://en.wikipedia.org/wiki/Near-field_communication), a short-range wireless communication, operates only within 10 cm, the beacon can operate at distances up to

50 m. Using beacon technology, data such as discount coupons and detailed descriptions can be transmitted to smartphones of customers visiting shopping centres, restaurants, museums, art galleries, movie theatres, and baseball fields.

Figure 3 Beacon data format (see online version for colours)



- **UUID:** a proximity UUID consisting of 16 bytes, usually a universal unique ID (UUID), is a unique ID for the product and may be configured as a UUID, depending on the service.
- **major and minor ID:** major ID and minor ID are 2 bytes each and can be set from 0 to 65,535. It is divided into major for the service group or region and minor value for the division within the same region.
- **Tx power:** Tx power (level) values indicates the received signal strength indicator (RSSI) (https://en.wikipedia.org/wiki/Received_signal_strength_indication) value measured by the beacon at 1 m. It is used to calculate the distance based on the measured RSSI value.

3 System architecture and functions

The implemented system consists of a smart parking sensor with communication function with the server as shown in Figure 4, a server to manage parking information, a Bluetooth beacon installed at each parking floor, and a smartphone app. Figure 5 is the overall system flow diagram. Each sensor transmits the parking space information to the server in real time via the Arduino system, which controls every ultrasonic sensors. When a vehicle driver executes the smartphone application and enters the car into the parking lot, the smartphone app detects the Bluetooth beacon, requests the parking information of specific floor from the server, and the parking space information is displayed in a map style.

Figure 4 System architecture (see online version for colours)



Figure 5 System flow (see online version for colours)

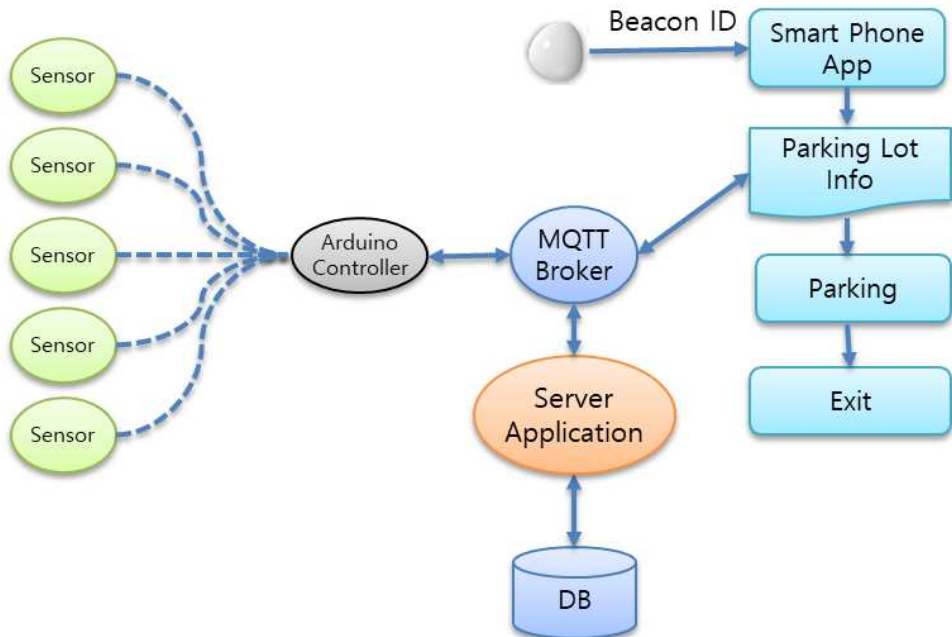
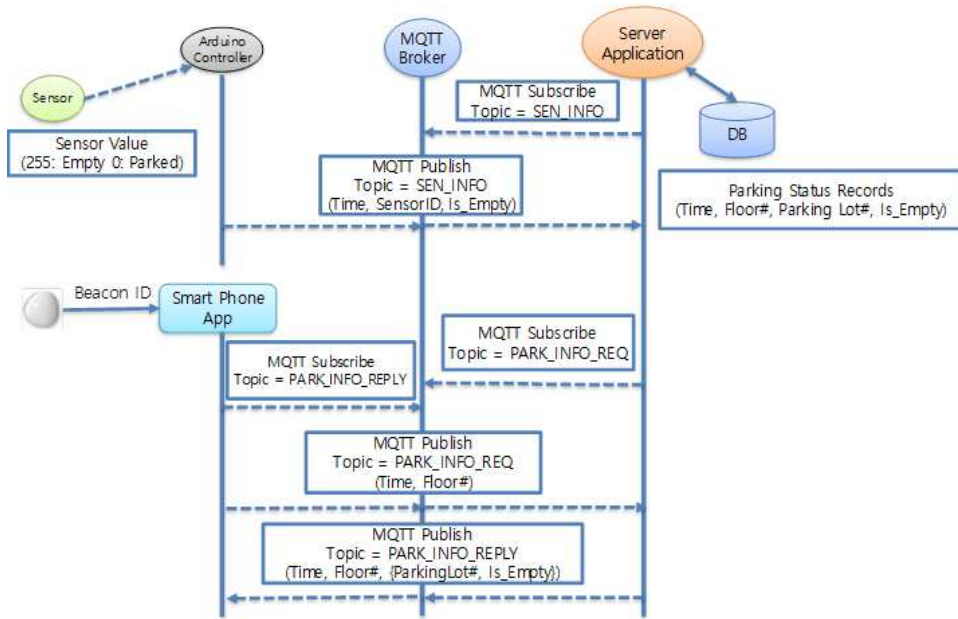


Figure 6 Communication and data structure (see online version for colours)

3.1 Communication procedure and data structure

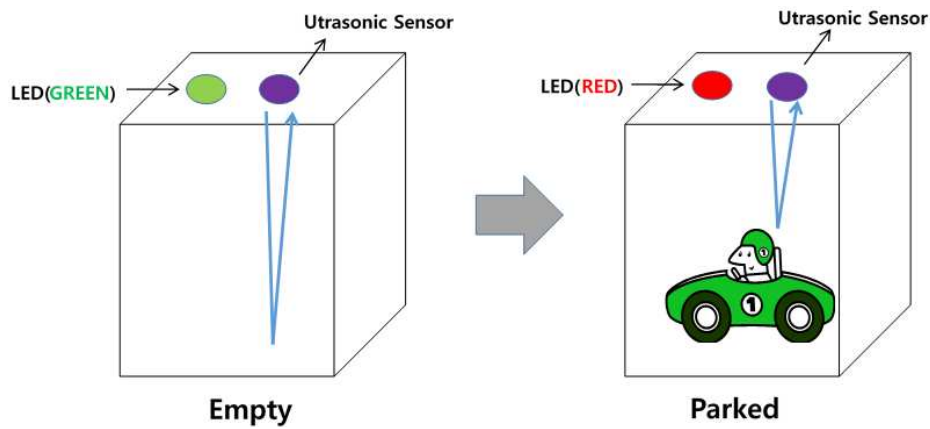
Figure 6 shows the communication procedure and data structure using MQTT protocol. Sensor information of each parking space is transmitted to server application using MQTT publish method with SEN_INFO topic and the data structure is (Time, Sensor ID, Is_Empty). Server application subscribes MQTT broker with SEN_INFO topic and save the information published by sensor controller into database. The data structures stored in the database is (Time, Floor #, ParkingLot #, Is_Empty). The smartphone application requests the server with the Publish method with PARK_INFO_REQ topic (Time, Floor #), and the server application publishes parking information (Time, Floor #, {ParkingLot #, Is_Empty}), with PARK_INFO_REPLY topic.

3.2 Parking sensor

The parking sensor uses serial line to communicate with Arduino controller. The Arduino board uses TCP communication module and MQTT protocol for communicating with the server. The controller continuously determines whether the parking space is occupied by checking ultrasonic sensor data value, and transmits the measurement data to the server. If parking space is empty, the value is high, otherwise the value is low because reflection distance is reduced. The data value of the sensor is transmitted to the server when the car is parked in an empty space or when the car is out of the parking space. When the data value of the sensor connected to the Arduino is updated, the LED lamp will also change colours depending on whether the space is occupied or not. Green means parking space is empty, while red indicates it is occupied. In this research, the sensor is designed to transmit data to the server using a wired TCP connection. Although wired Ethernet shield

is used for sample implementation in this paper, a wireless communication can be applied. Figure 7 shows how the parking sensor works.

Figure 7 Parking sensor (see online version for colours)



3.3 Bluetooth beacon

In this research, the smartphone app is designed to communicate with Bluetooth beacon installed at the entrance of each parking floor. The smartphone reads the Beacon ID and sends it to the server with MQTT protocol. The server transmits parking information on the floor of specified ID to the smartphone also with MQTT publishing protocol. Figure 8 and 9 show the installation location of the beacon and how it works.

Figure 8 Bluetooth beacon at parking lot entrance (see online version for colours)

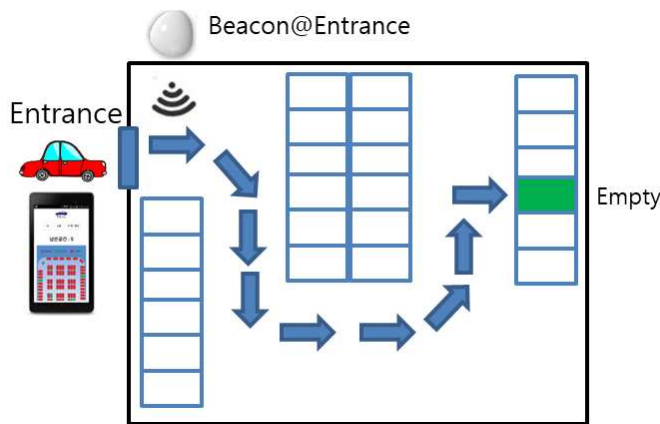


Figure 9 Bluetooth beacon for location identification (see online version for colours)

3.4 Android app

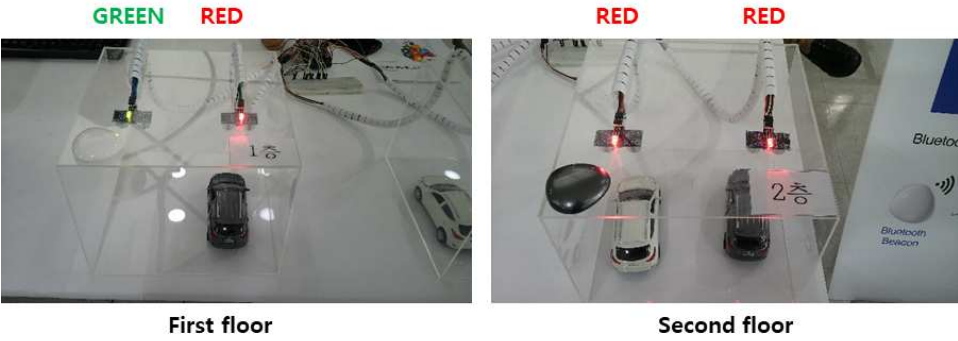
A driver's smartphone app detects the signal of the Bluetooth beacon on the floor, communicates with the server, and displays real-time parking information on the screen. It communicates with server using MQTT protocol. The driver executes the smartphone app and activates the Bluetooth function before entering the parking lot. Figure 10 is an execution screen of the parking guide application. As shown on the screen, the app displays the parking information of all floors at first and when it detects a Beacon signal, it displays parking space information of the place where the car is currently located. Green means a parking space is available and red is not.

Figure 10 Smart parking guidance app (see online version for colours)

4 Implementation and experiment

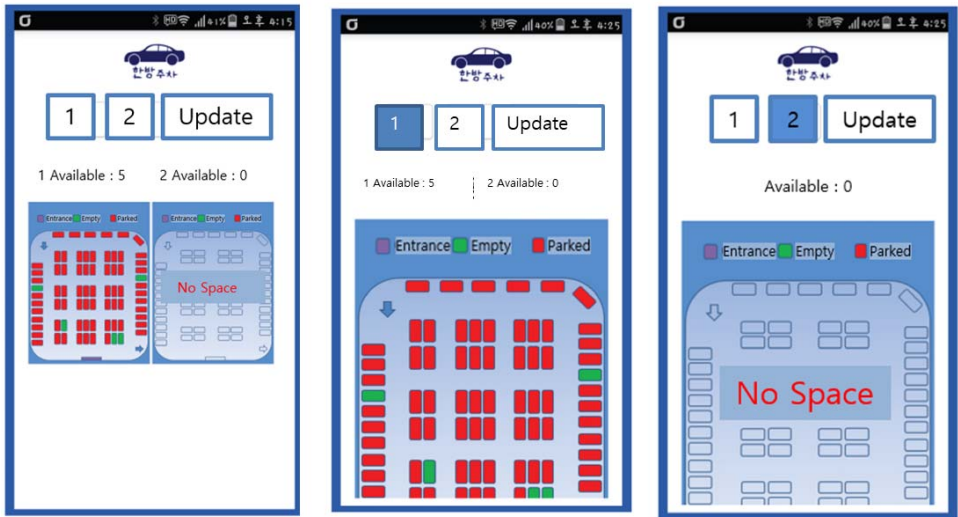
A sample parking lot was built as shown in Figure 11. In Figure 11, the parking space on the first floor shows a GREEN LED shows the empty space and a RED LED where the vehicle is parked. On the other hand, no space is available on second floor and therefore all LEDs are RED. Figure 11 shows first floor and second floor parking map in which there are available parking space in first floor and no space on second floor.

Figure 11 Parking lot sample (see online version for colours)



As shown in Figure 12, assume that a car enters parking lot #1 (first floor), the app automatically displays parking map of first floor and it shows there are 5 available parking spaces. If a car enters parking lot #2 (second floor), as experimented in Figure 11, since there is no parking space available, the app displays “No Space” for the second floor.

Figure 12 Parking guidance app (see online version for colours)

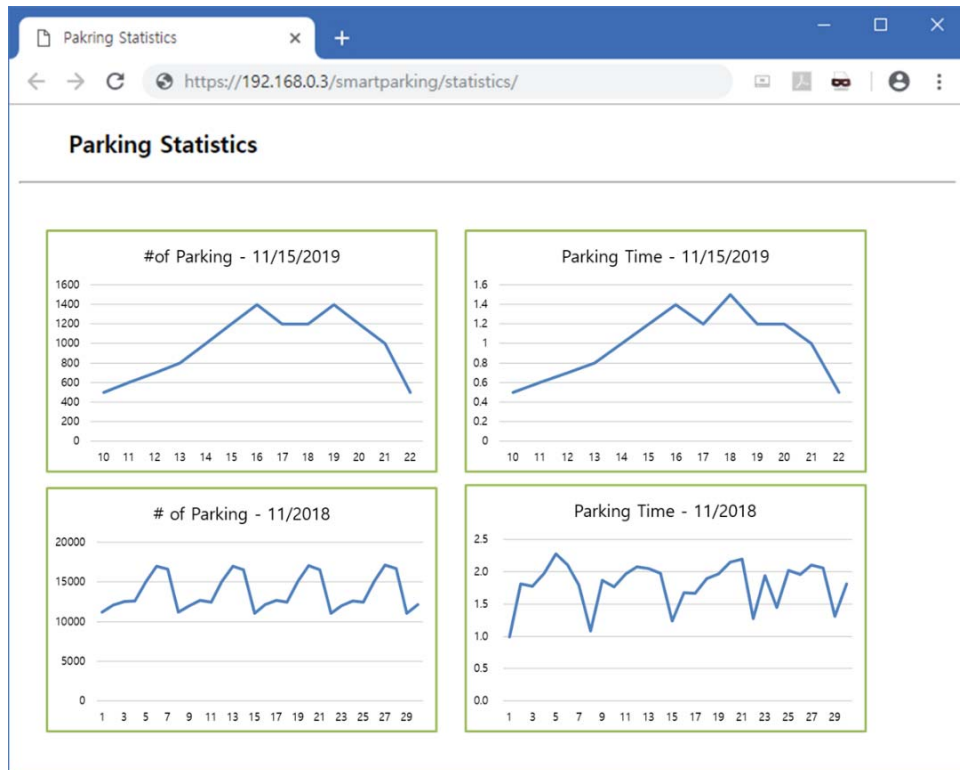


5 Parking statistics display

In this paper, on the parking management server, the author developed a web view page to display parking statistical information. The parking data in database is consisted of (*time, floor number, parking space number, parking flag*). In terms of marketing, various meaningful statistical information can be generated by processing parking data stored in the database. Currently implemented are # of parking and average parking time. The two types of information can be displayed as hourly, daily, and monthly basis. Figure 13

shows an example of parking statistics data output screen. It shows one day parking statistics on time basis and daily average values during a month. In addition, various types are also implemented, such as weekly report, yearly report, etc.

Figure 13 Parking statistics web view (see online version for colours)



6 Conclusions and future research

In this paper, IOT-based smart parking management system is implemented using MQTT protocol. The implemented parking guidance system uses Bluetooth beacon to display the location-based parking information in real time on the smartphone app so that the driver can check the parking space quickly. The IoT sensors are controlled by an Arduino board and the data is transmitted to the server using the MQTT protocol, which is one of the standard IoT protocol. The server stores the received sensor data through by means of an MQTT topic and provides the real-time parking status to the driver. The stored parking data can be used as meaningful statistical information such as the number of parking cars and average parking time, which means a shopping time in terms of stores. The smart parking management system implemented in this paper can be utilised as a tool for customer service in a commercial facility having a large parking facility such as a large shopping mall and can be utilised as information for customer management required for marketing based on parking data.

Future research to follow in this study is to install Bluetooth beacon in various places inside the parking lot so that the smartphone app automatically stores the parking position and adds the audio parking guide function so that the driver can park the car without seeing the parking map. In addition, the author will study intelligent customer management technique to produce meaningful information through big data analysis by combining stored parking information and marketing information.

Acknowledgements

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